

Reviewer Pack — Minimal Local Induction Ring Rank and Communication Complexi...

Entry 6452e27bd56c · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Induction Ring Rank and Communication Complexity Rank Inequality

Entry ID: 6452e27bd56c **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-05 21:37:00 UTC

1. Conjecture

Field A (mathematical branch): Algebra (Local Induction Rings) **Field B** (complexity object): Communication Complexity (Matrix Rank)

Statement:

For every CNF φ with n variables, the minimal local induction ring rank of its clause-indicator polynomial modulo 2 is lower-bounded by a linear function of the communication complexity rank of the corresponding constraint satisfaction problem, i.e., $m_{\text{lir}}(\varphi) = \Theta(r_{\Gamma}(\varphi))$ for some constant c .

Rationale (proposer's reasoning):

The use of local induction rings in algebra can provide an invariant that captures the structure of Boolean functions. This conjecture posits a direct link between this algebraic structure and the communication complexity, which is a central topic in complexity theory. If true, it could potentially lead to new algorithms for solving constraint satisfaction problems.

Taxonomy category: local_induction_rings (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 23ada529b7b884ee

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a CNF φ with $n \leq 40$ variables, if $m_{\text{lir}}(\varphi) = \Theta(r_{\Gamma}(\varphi))$ is supported by at least 80% of seeds and the correlation coefficient between $m_{\text{lir}}(\varphi)$ and $r_{\Gamma}(\varphi)$ is ≥ 0.9 .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	SAFE

4. Novelty audit

Verdict: ADJACENT_OK against 8 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Minimal Local Induction Ring Rank AND Communication Complexity Rank
- Clause-Indicator Polynomial MOD 2 AND Constraint Satisfaction Problem Communication Complexity - Local Induction Rings Matrix Rank Inequality Communication Complexity

Top relevant hits considered: - [http://arxiv.org/abs/1403.8106v1] Recent advances on the log-rank conjecture in communication complexity - [http://arxiv.org/abs/2311.01864v1] SortNet: Learning To Rank By a Neural-Based Sorting Algorithm - [http://arxiv.org/abs/2510.02583v2] The Log-Rank Conjecture: New Equivalent Formulations - [http://arxiv.org/abs/1109.5789v1] Approximation Complexity of Complex-Weighted Degree-Two Counting Constraint Satisfaction Problems - [http://arxiv.org/abs/1503.07310v5] The Complexity of Phylogeny Constraint Satisfaction Problems - [http://arxiv.org/abs/cs/0503082v1] Spines of Random Constraint Satisfaction Problems: Definition and Connection with Computational Complexity - [http://arxiv.org/abs/1401.4438v1] Integral closure of rings of integer-valued polynomials on algebras - [http://arxiv.org/abs/1102.2932v2] Applications of Monotone Rank to Complexity Theory

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=3.2s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

# Helper functions for Gaussian elimination and matrix operations
def gaussian_elimination(matrix):
    n = len(matrix)
    for i in range(n):
        # Find pivot row
        max_row = i
        for k in range(i + 1, n):
            if abs(matrix[k][i]) > abs(matrix[max_row][i]):
                max_row = k
        matrix[i], matrix[max_row] = matrix[max_row], matrix[i]

        # Eliminate below pivot
        for k in range(i + 1, n):
            factor = Fraction(matrix[k][i], matrix[i][i])
            for j in range(n):
                matrix[k][j] -= factor * matrix[i][j]

    # Back substitution
    x = [0] * n
```

```

    for i in range(n - 1, -1, -1):
        x[i] = Fraction(matrix[i][-1], matrix[i][i])
        for j in range(i + 1, n):
            x[i] -= x[j] * matrix[i][j]

    return x

def matrix_multiply(A, B):
    m = len(A)
    k = len(B)
    n = len(B[0])
    C = [[0] * n for _ in range(m)]
    for i in range(m):
        for j in range(n):
            for p in range(k):
                C[i][j] += A[i][p] * B[p][j]
    return C

def matrix_power(A, n):
    result = [[Fraction(1) if i == j else Fraction(0) for j in range(len(A))] for i in range(len(A))]
    while n > 0:
        if n % 2 == 1:
            result = matrix_multiply(result, A)
            A = matrix_multiply(A, A)
            n //= 2
    return result

def communication_complexity(phi):
    # Placeholder function to compute communication complexity rank  $r_\Gamma(\phi)$ 
    # This is a dummy implementation and should be replaced with actual logic.
    return random.randint(1, 10)

# Function to generate a random CNF with n variables
def generate_cnf(n):
    cnf = []
    for _ in range(random.randint(5, 10)):
        clause = [random.choice([-1, 1]) * (i + 1) for i in range(n)]
        if len(set(clause)) == len(clause): # Ensure no duplicate literals
            cnf.append(clause)
    return cnf

# Function to compute the minimal local induction ring rank  $m_{\text{lir}}(\phi)$ 
def minimal_local_induction_ring_rank(cnf):
    n = max(abs(lit) for clause in cnf for lit in clause)
    poly = [0] * (2 ** n)
    for clause in cnf:
        product = 1
        for lit in clause:
            product *= (-1 if lit < 0 else 1) * (1 - Fraction(1, 2) ** abs(lit))
        poly[sum(abs(lit) for lit in clause)] += product

    matrix = [[poly[i ^ (1 << j)] for j in range(n)] for i in range(2 ** n)]
    rank = sum(1 for row in gaussian_elimination(matrix) if any(row))
    return rank

# Main function to run a single trial

```

```

def run_trial(seed: int) -> dict:
    random.seed(seed)

    results = []
    for _ in range(30):
        n = random.choice([5, 10, 15, 20, 30, 40])
        phi = generate_cnf(n)

        m_lir = minimal_local_induction_ring_rank(phi)
        r_Γ = communication_complexity(phi)

        results.append((m_lir, r_Γ))

    if not results:
        return {
            "metric_name": "minimal_local_induction_ring_rank",
            "metric_value": 0,
            "instances_tested": 0,
            "n_max": 0,
            "conjecture_holds": False,
            "counterexample": "empty_results"
        }

    m_lir_values = [m for m, _ in results]
    r_Γ_values = [r for _, r in results]

    correlation_coefficient = sum((m - sum(m_lir_values) / len(m_lir_values)) * (r - sum(r_Γ_values) / len(r_Γ_values))) / len(m_lir_values)
    correlation_coefficient /= math.sqrt(sum((m - sum(m_lir_values) / len(m_lir_values)) ** 2 for m in m_lir_values) / len(m_lir_values))

    conjecture_holds = correlation_coefficient >= 0.9
    counterexample = "" if conjecture_holds else "correlation_coefficient=<{}>".format(correlation_coefficient)

    return {
        "metric_name": "minimal_local_induction_ring_rank",
        "metric_value": sum(m_lir_values) / len(m_lir_values),
        "instances_tested": 30,
        "n_max": max(n for _, n in results),
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

# Main execution block
if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print("TRIAL: {}".format(trial_result))
        results.append(trial_result)

    if all(result["conjecture_holds"] for result in results):
        mean_value = sum(result["metric_value"] for result in results) / len(results)
        support_fraction = 1.0
        print("RESULT: SUPPORTED mean={} std=0 support_fraction={}".format(mean_value, 0, support_fraction))
    else:

```

```

first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["con]
counterexample = results[seeds.index(first_failing_seed)][["counterexample"]]
print("RESULT: FALSIFIED counterexample={} first_failing_seed={}".format(counterexample, f

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_163fc8dc.py", line 140, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_163fc8dc.py", line 101, in run_trial
    m_lir = minimal_local_induction_ring_rank(phi)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_163fc8dc.py", line 89, in minimal_local
    rank = sum(1 for row in gaussian_elimination(matrix) if any(row))
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_163fc8dc.py", line 33, in gaussian_elim
    matrix[k][j] -= factor * matrix[i][j]
    ~~~~~^~~~
IndexError: list index out of range

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed during execution, which prevents us from verifying the conjecture's validity. | next: Re-examine the code for potential bugs and ensure it can run to completion without errors.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	18686
2	propose	ollama_remote	glm4:latest	0	0	12746
3	propose	ollama_remote	glm4:latest	0	0	13892
4	preregistration	ollama_remote	glm4:latest	0	0	9214
5	novelty	ollama_remote	glm4:latest	0	0	8392
6	novelty	ollama_remote	glm4:latest	0	0	10480
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	29376

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10971
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15100
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16852
11	judge	ollama_remote	glm4:latest	0	0	15339

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 161048 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in research/audit/6452e27bd56c.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/6452e27bd56c.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/6452e27bd56c.tar.gz (if generated)